

27	B	$P = \frac{mc\Delta\theta}{t} \Rightarrow t = \frac{mc\Delta\theta}{P}$ For steady d.c $t = \frac{mc\Delta\theta}{I^2 R} \text{ as } P = I^2 R$ For a.c with resistance 2R, $t_{ac} = \frac{mc\Delta\theta}{\left(\frac{I}{\sqrt{2}}\right)^2 2R} = \frac{mc\Delta\theta}{I^2 R} \text{ as } P_{ac} = \left(\frac{I}{\sqrt{2}}\right)^2 2R$ $t_{ac} = T$
28	B	Current = power / voltage = $1 \times 10^6 / 66 \times 10^3 = 15.15 \text{ A}$